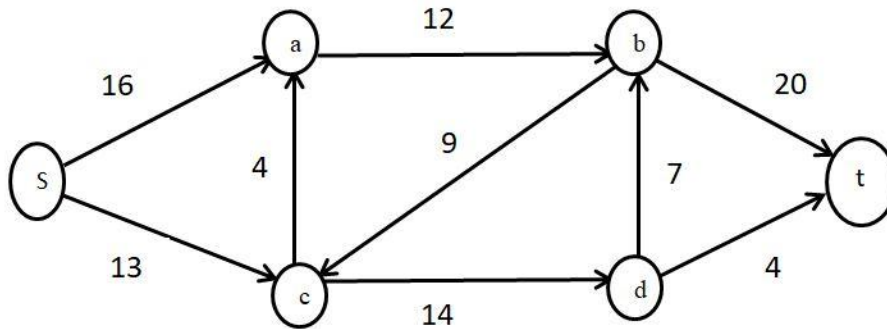
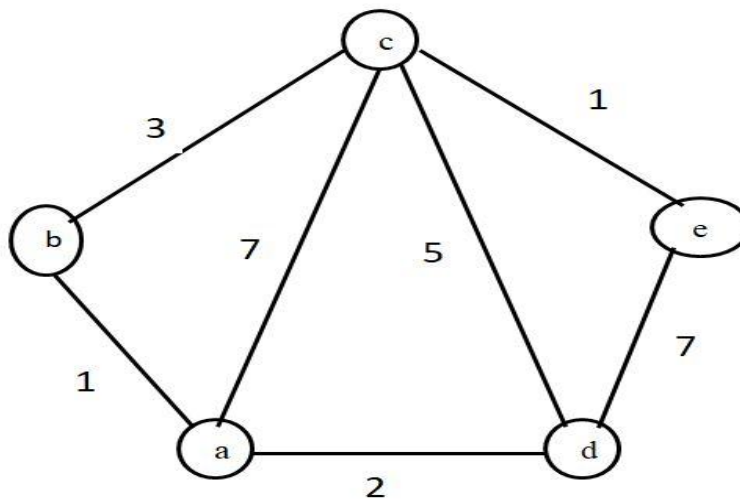


Questions Bank Unit-4, 5

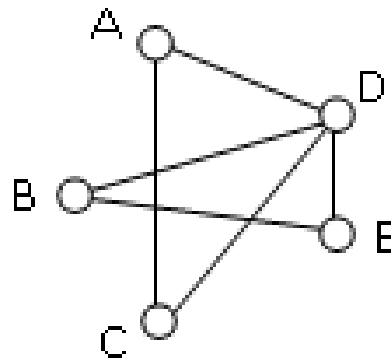
- Find the maximum flow in the transport network in the following figure and determine the corresponding minimum cut.



- Find the shortest distance and shortest path from 'a' to 'e' in the given weighted graph using Dijkstra algorithm.

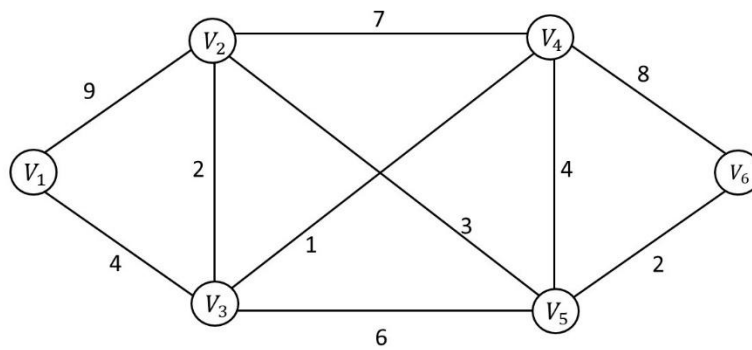


- Define Euler circuit. Use Fleury's algorithm to find the Euler Circuit in the following graph

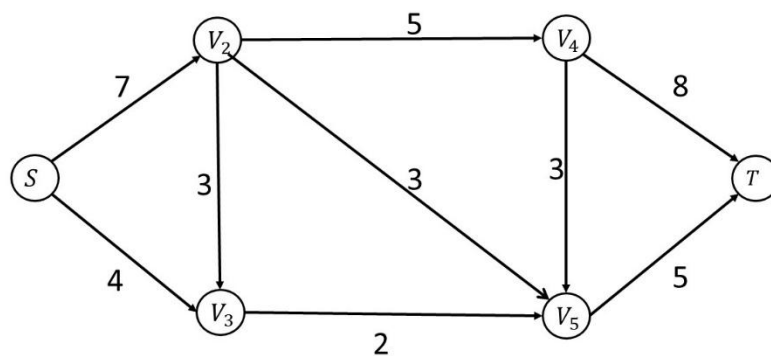


- Prove that for a connected simple planar graph has a vertex degree that does not exceed 5.

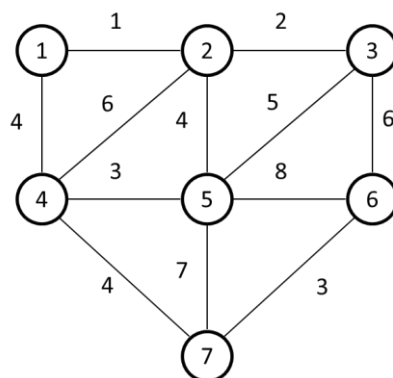
5. Use Dijkstra's algorithm to find the shortest path between the vertex V_1 and V_6



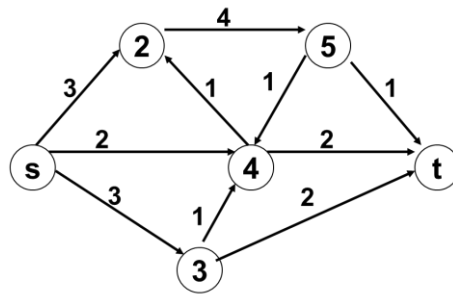
6. Find the maximum flow in the following network using the Ford-Fulkerson algorithm



7. Use Kruskal's algorithm find a minimal spanning tree (MST) of the following graph.



8. Find the maximal flow of the following network using Ford-Fulkerson algorithm



9. Prove that Modulo 5 under addition ($\mathbb{Z}_5, +$) forms a group. Further prove that the group is cyclic.
10. What is a group code? Given the set of codewords: $\{000, 111\}$, check whether this set forms a group under bitwise XOR.
11. State Lagrange's Theorem.
12. If G is a group of order 10 and H is a subgroup of order 5, how many distinct left cosets does H have in G ?
13. Construct a binary linear code using a generator matrix and explain how syndrome decoding can be used to correct errors.
14. Show that $(G, \oplus_8)$ is an abelian group, where $G = \{0,1,2,3,4,5,6,7\}$.